

PREFACE

IPU (Intelligent Processing Unit) is an architectural concept that explores an alternative approach to building intelligent computing systems.

This document does not define a specific artificial intelligence model, learning algorithm, memory implementation, hardware technology, or software framework.

Instead, it defines the architectural principles, interface contracts, and communication model that allow independent components to operate together as a unified intelligent system.

Throughout this document, implementation details are intentionally left unspecified.

Any technology, model, algorithm, or hardware platform capable of fulfilling the defined interface contracts may be integrated into the IPU architecture.

The purpose of this document is not to prescribe how intelligence must be built.

Its purpose is to define how independently developed intelligent components can communicate, cooperate, and evolve within a common architectural framework.

Document Scope

This document focuses on architectural design.

It describes:

- System architecture
- Module responsibilities
- Interface contracts
- Communication principles
- Data exchange
- Design philosophy

This document intentionally does not define:

- AI model architecture
- Machine learning algorithms
- Training methods
- Hardware implementation
- Memory technology
- Programming language
- Database systems
- Operating systems

Design Philosophy

The IPU architecture follows one fundamental principle:

Focus on Interface, Not Implementation.

Each module defines what it is responsible for,
how it communicates,
and what information it exchanges.

How the module performs its internal computation
is intentionally left open.

This separation allows independent evolution of hardware,
software, AI models,
memory systems,
and future technologies
without requiring architectural redesign.

Reading Guide

Volume I

Introduces the original IPU architectural concept.

Volume II

Describes one possible implementation strategy using today's technology.

Neither volume specifies mandatory implementation details.

All implementation examples are provided solely to illustrate architectural compatibility.

Introduction

The development of artificial intelligence has progressed rapidly over the past decades.

Large language models, machine learning systems, robotics, distributed computing, and advanced memory technologies have significantly expanded the capabilities of modern intelligent systems.

Despite these advances, most existing systems are still designed around specific implementations.

The architecture often becomes tightly coupled with a particular AI model, hardware platform, memory technology, or software framework.

As these technologies continue to evolve, replacing or upgrading one component frequently requires modifications across multiple parts of the system.

This document explores an alternative architectural approach.

Instead of defining how intelligence should be implemented, the IPU architecture defines how independently developed intelligent components should communicate and cooperate.

By separating architectural contracts from implementation details, the system becomes open to continuous technological evolution without requiring fundamental architectural redesign.

The objective of IPU is not to replace existing artificial intelligence technologies.

Its objective is to provide a stable architectural foundation upon which different generations of intelligent technologies may coexist, evolve, and integrate over time.

Why IPU?

Technology evolves continuously.

Artificial intelligence models improve.

Hardware architectures change.

Memory technologies advance.

Communication protocols become faster.

Software frameworks are replaced.

However, architectural principles should remain stable.

The IPU architecture is based on the idea that intelligence should not depend on a specific implementation.

Instead, every subsystem communicates through well-defined interface contracts.

As long as those contracts are preserved, individual modules may evolve independently without affecting the overall architecture.

This approach encourages modularity, interoperability, maintainability, and long-term scalability.

Core Philosophy

The IPU architecture is built upon several fundamental principles.

- Architecture before implementation.
- Interface before algorithm.
- Responsibility before optimization.
- Communication before integration.
- Open by design.
- Modular by default.
- Technology agnostic.
- Implementation independent.

The architecture intentionally avoids prescribing specific technologies.

Instead, it provides a common language through which independent systems can cooperate.

How to Read This Document

This document is organized into two complementary volumes.

Volume I introduces the original architectural concepts of the Intelligent Processing Unit (IPU).

Its purpose is to explain the architectural vision without assuming any specific implementation technology.

Volume II presents one possible implementation strategy using technologies that are available today.

It demonstrates how the architectural principles introduced in Volume I may be applied in practice while intentionally leaving implementation details open.

Readers are encouraged to study Volume I before proceeding to Volume II, as the second volume assumes familiarity with the architectural concepts introduced in the first.

VOLUME I

Original Architecture Concept

Understanding the architectural vision before implementation.

Chapter 01

What is IPU?

The Intelligent Processing Unit (IPU) is an architectural concept for designing intelligent computing systems.

Unlike traditional processor architectures that primarily focus on computation, the IPU architecture is designed around the interaction of intelligence, memory, communication, goals, feedback, and continuous adaptation.

Rather than defining a single processing core, IPU describes a collection of cooperating architectural components that work together through standardized interfaces.

Each component has clearly defined responsibilities while remaining independent from the technologies used to implement it.

The architecture intentionally separates system behavior from implementation details.

As a result, different hardware platforms, artificial intelligence models, memory technologies, communication mechanisms, and future innovations may be integrated without fundamentally changing the architectural design.

Design Objective

The objective of IPU is to establish a common architectural framework for intelligent systems.

The architecture focuses on:

- Modular system design
- Independent subsystem evolution
- Standardized interface contracts
- Flexible technology integration
- Long-term architectural stability

The architecture does not attempt to define intelligence itself.

Instead, it defines how intelligent components cooperate within a unified system.

What IPU Is Not

IPU is not an artificial intelligence model.

IPU is not a language model.

IPU is not a machine learning algorithm.

IPU is not a hardware implementation.

IPU is not an operating system.

IPU is not a programming framework.

Instead, IPU is an architectural framework that defines how independent intelligent components communicate and cooperate.

Chapter 02

Why an Architectural Approach?

Modern intelligent systems continue to become more capable as artificial intelligence technologies rapidly evolve.

New AI models, memory systems, hardware platforms, and software frameworks are introduced continuously.

While these technologies improve individual capabilities, they often introduce architectural fragmentation.

Many intelligent systems become tightly coupled to specific implementations, making future integration and evolution increasingly difficult.

Replacing one major component may require modifications across multiple parts of the system.

As intelligent systems continue to grow in complexity, architectural consistency becomes increasingly important.

The Challenge

The primary challenge is not the lack of intelligent algorithms.

The challenge is enabling independently evolving intelligent components to cooperate within a common architectural framework.

Different technologies should be able to communicate without requiring the entire system to be redesigned whenever one component changes.

This challenge becomes increasingly significant as artificial intelligence technologies continue to evolve.

The IPU Perspective

The IPU architecture approaches this challenge by defining responsibilities and communication interfaces instead of implementation details.

Each architectural component is responsible for a specific function within the system.

How that function is implemented is intentionally left open.

This separation allows different technologies to evolve independently while maintaining architectural compatibility through stable interface contracts.

Chapter 03

Architectural Overview

The Intelligent Processing Unit (IPU) is organized as a collection of cooperating architectural components.

Each component is responsible for a specific aspect of intelligent system behavior.

Rather than relying on a single centralized processing unit, the architecture distributes responsibilities across multiple specialized components that communicate through well-defined interfaces.

Together, these components form a unified architectural framework capable of supporting intelligent computation, communication, memory interaction, planning, and continuous adaptation.

Architectural Components

The IPU architecture consists of multiple architectural components.

Each component performs a distinct responsibility within the overall system.

Although implementations may differ, the architectural responsibilities remain consistent.

The primary architectural components include:

- Input System
- Conversational Engine
- Memory System
- Goal Manager
- Planning and Reasoning
- Internal State
- Action System
- Feedback Network
- System Interface

Cooperation Instead of Centralization

Traditional processor architectures are often organized around a central processing unit responsible for most computational activities.

The IPU architecture follows a different architectural philosophy.

Instead of concentrating responsibilities within a single processing element, intelligent behavior emerges through cooperation among specialized architectural components.

Each component focuses on its own responsibility while exchanging information through standardized communication interfaces.

This architectural separation improves modularity, maintainability, and implementation flexibility without prescribing how each component must be built.

Architectural Independence

Each architectural component is defined by its responsibility rather than its implementation.

The architecture specifies what information a component receives, what responsibilities it performs, and what information it produces.

How those responsibilities are implemented is intentionally left outside the scope of the architecture.

This separation allows implementations to evolve while preserving architectural compatibility.

Chapter 04

Architectural Flow

The IPU architecture is designed as a continuous flow of information rather than a sequence of isolated computations.

Information enters the system through one or more input interfaces.

As information moves through the architecture, multiple components cooperate by interpreting, storing, evaluating, planning, and generating appropriate responses.

Throughout this process, memory, goals, internal state, and feedback continuously influence system behavior.

The architecture therefore represents an adaptive processing flow instead of a fixed execution pipeline.

High-Level Information Flow

Although implementations may vary, the architectural flow generally follows these stages.

1. Information is received through the Input System.
2. The Conversational Engine interprets incoming information.
3. Relevant information may be retrieved from the Memory System.
4. The Goal Manager determines whether the information relates to current objectives.
5. Planning and Reasoning evaluate possible actions.
6. Internal State provides contextual conditions that may influence decision making.
7. The Action System generates responses or actions.
8. Feedback Network evaluates outcomes and supports future adaptation.

This sequence represents a conceptual architectural flow rather than a mandatory execution order.

Continuous Feedback

Unlike traditional sequential processing models, the IPU architecture does not assume that processing ends after an action is produced.

Outputs generated by the system may become new inputs.

Internal state may change continuously.

Memory may be updated throughout execution.

Goals may evolve.

Feedback continuously influences future processing.

As a result, the architecture supports continuous operation rather than isolated request-response execution.

Architectural Flexibility

The architecture intentionally avoids defining strict execution timing or synchronization mechanisms.

Architectural components may operate sequentially, concurrently, asynchronously, or through other implementation-specific approaches.

Only the architectural responsibilities and communication interfaces are defined.

Execution models remain implementation dependent.

Chapter 05

Architectural Components

Every architectural component within the IPU is designed to fulfill a specific responsibility.

Rather than combining multiple functions into a single processing unit, the architecture separates responsibilities into independent components.

Each component communicates with other components through standardized interfaces while remaining independent of implementation technology.

This architectural separation allows the system to evolve without requiring every component to change simultaneously.

Responsibility-Oriented Design

The IPU architecture assigns responsibilities rather than algorithms.

Each component is defined by:

- What information it receives.
- What responsibility it performs.
- What information it produces.

The architecture intentionally avoids specifying how these responsibilities must be implemented.

Implementation remains the responsibility of future system designers.

Primary Components

The current IPU architectural concept includes the following primary components.

- Input System
- Conversational Engine
- Memory System
- Goal Manager
- Planning and Reasoning
- Internal State
- Action System
- Feedback Network
- System Interface

Each component will be introduced individually in the following chapters.

Independent Evolution

One of the primary objectives of the IPU architecture is to support independent evolution.

A component may improve internally without requiring architectural changes to unrelated components.

As long as architectural responsibilities and communication interfaces remain compatible, implementations may continue evolving over time.

This principle encourages long-term maintainability and technology independence.

Chapter 06

Input System

The Input System serves as the primary entry point of information into the IPU architecture.

Its responsibility is to receive information from external or internal sources and present it to the architecture through standardized interfaces.

The architecture does not restrict the type of input.

Inputs may originate from sensors, users, software systems, networks, memory recall, or future technologies.

The Input System is responsible only for information acquisition.

Interpretation of information is performed by other architectural components.

Chapter 07

Conversational Engine

The Conversational Engine is responsible for interpreting incoming information and transforming it into meaningful internal representations.

Its responsibility is communication and understanding rather than long-term storage or decision making.

The architecture intentionally does not define how understanding is achieved.

Any implementation capable of fulfilling the required architectural responsibilities may be integrated.

Chapter 08

Memory System

The Memory System is responsible for storing, retrieving, organizing, and providing information required by the architecture.

Memory is treated as an active architectural component rather than passive storage.

Architectural components may request information from memory whenever required.

The architecture does not prescribe specific memory technologies or storage mechanisms.

Chapter 09

Goal Manager

The Goal Manager maintains the objectives that guide system behavior.

Goals influence planning, decision making, prioritization, and action selection.

Goals may originate from users, higher-level systems, internal planning, or future architectural extensions.

The architecture defines the responsibility of goal management without prescribing implementation methods.

Chapter 10

Planning, Action, Feedback, and Internal State

Planning and Reasoning evaluate possible actions based on available information, memory, goals, and current context.

Internal State represents the current condition of the system that may influence future behavior.

The Action System produces external or internal actions according to planning decisions.

The Feedback Network continuously evaluates outcomes and provides information that may influence future processing.

Together, these architectural components support continuous adaptation while remaining implementation independent.

Volume I introduces the architectural concepts of the Intelligent Processing Unit.

The purpose of this volume is to describe the architectural vision and the responsibilities of each component without prescribing implementation details.

The following volume presents one possible implementation approach using technologies that are available today.

VOLUME II

First Implementation Using Today's Technology

One possible implementation strategy.
Not a mandatory implementation.

Chapter 11

The First IPU Prototype

The architectural concepts introduced in Volume I intentionally avoid prescribing implementation technologies.

This volume presents one possible implementation strategy using technologies that are widely available today.

The purpose is not to define the final implementation of IPU.

Instead, it demonstrates how the architectural principles may be translated into a practical prototype while preserving implementation independence.

The implementation presented in this volume should be regarded as an example rather than a requirement.

Chapter 12

Conversational Engine

In this implementation example, the Conversational Engine serves as the primary interface for natural language understanding and communication.

Its responsibility is to interpret incoming information and produce meaningful internal representations that may be used by other architectural components.

The IPU architecture does not require any specific artificial intelligence model.

Large Language Models, custom AI systems, rule-based systems, or future technologies may all fulfill this responsibility provided that they satisfy the architectural interface requirements.

Chapter 13

Memory Interface

The Memory System communicates with other architectural components through standardized interfaces.

Architectural components request information without requiring knowledge of the underlying storage technology.

Whether memory is implemented using databases, vector stores, knowledge graphs, distributed storage, or future memory technologies is intentionally left outside the architectural specification.

Only the communication interface is considered part of the architecture.

Chapter 14

Goal and Planning Interface

Goals and planning are exposed through architectural interfaces rather than implementation-specific algorithms.

Different planning techniques may therefore be integrated without modifying the surrounding architecture.

This separation enables continuous improvement of planning capabilities while preserving architectural compatibility.

The architecture specifies responsibilities rather than planning methodologies.

Chapter 15

Sensory Interface

The Input System may receive information from multiple independent sources.

These sources may include users, software services, sensors, external devices, digital communication channels, or future technologies.

The architecture standardizes how information enters the system while intentionally leaving input technologies unrestricted.

This approach allows new input mechanisms to be integrated without redesigning the architecture.

Chapter 16

Internal State Interface

The Internal State Interface provides architectural access to the current operational condition of the system.

Internal state may include contextual information, execution status, resource conditions, temporary variables, or other implementation-specific information.

The architecture defines how components exchange state information without prescribing how that information is internally represented or maintained.

This separation allows different implementations to manage internal state according to their own design requirements.

Chapter 17

Action System Interface

The Action System is responsible for transforming architectural decisions into observable outputs.

Actions may include natural language responses, digital operations, external device control, memory updates, communication with other systems, or future implementation-specific capabilities.

The architecture specifies the communication interface between decision making and action execution while remaining independent of the underlying execution technology.

Chapter 18

Feedback Interface

The Feedback Network enables architectural components to receive information about previous actions and current system behavior.

Feedback supports continuous adaptation by providing information that may influence future planning, memory updates, goal evaluation, and internal state.

The architecture defines the existence of feedback communication without restricting how feedback is generated, evaluated, or utilized.

Chapter 19

Architectural Integration

The implementation presented throughout this volume demonstrates that the IPU architecture may be constructed using existing technologies.

Individual implementation choices may differ significantly depending on application requirements.

Artificial intelligence models, memory technologies, communication protocols, databases, operating systems, and hardware platforms may all evolve independently while preserving architectural compatibility through standardized interfaces.

This implementation therefore represents only one possible realization of the architectural concepts introduced in Volume I.

Chapter 20

Conclusion

The Intelligent Processing Unit (IPU) presented in this document is an architectural concept for intelligent computing systems.

Rather than prescribing specific technologies, algorithms, or hardware designs, the architecture defines responsibilities, communication interfaces, and cooperation between architectural components.

By separating architecture from implementation, the IPU architecture is intended to remain adaptable as intelligent technologies continue to evolve.

This document does not claim to present a complete or fully validated implementation.

Instead, it provides an architectural foundation intended to encourage discussion, experimentation, prototype development, and future architectural exploration.

The future of the IPU architecture depends not only on this document, but also on continued research, engineering validation, practical implementation, and contributions from future developers and researchers.